Emergency measures in underground environments

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This document presents the requirements for preparing and managing tunnel fires and can serve as a basis for the preparation of associated project documentation.

1. Fire Site Installation plan

Each project develops and maintains its Fire Site Installation Plan, including but not limited to:

- Extinguishing equipment available (fire hydrants, fire extinguishers, etc.)
- Storage of dangerous products (chemicals, flammable, gas cylinders)
- Firefighter and hydrant access available in the area
- The infirmary if applicable
- Emergency resources (Man injury basket, AED, first aid kits, etc.)
- Campsites, containers, etc.
- Muster points
- etc.

2. Tunnel Emergency Response Plan

The main objectives of the project/worksite Tunnel Emergency Response Plan are:

- To ensure the effective and efficient management of a potential fire in a tunnel
- Define internal and external roles and responsibilities
- Assess the situation and implement appropriate mitigation and control measures

Each project/construction site:

- Develops and maintains its own Tunnel Emergency Response Plan, which details the preparation and response measures to the various anticipated emergency situations (fire in particular)
- Develops and maintains an emergency management flowchart which summarizes the elements of the Emergency Plan (processes, key personnel, key contacts)
- Develops an annual plan for emergency exercises, carries them out, and updates its procedures if necessary based on drill feedback

The project/worksite Tunnel Emergency Response Plan takes into consideration all contractual requirements relating to emergency and crisis management, such as:

- The emergency management system of the Customer and partners
- The requirements of the Client and partners in the relationship with local authorities and media
- Local and legal requirements

For example, the Paris firefighters impose an opening to the outside (cross passage or shaft etc.) every 2000m maximum, based on the fact that beyond that, intervention on the part would be made almost impossible for the round-trip length of the trip. This data must be known upstream of the project because it has an impact on the planning of opening of cross passages, for example.

Its development is based on the evaluation:

- Risks associated with work activities
- Natural, geographic and environmental risks associated with workplaces
- Risks linked to the health of participants
- Security risks

The Emergency Plan is reviewed:

- Immediately after an incident or any emergency situation actually experienced
- Immediately after each exercise, in order to incorporate any improvement deemed necessary
- Following changes in site configuration, equipment and materials used, organization and key personnel.

3. Tunnel access control

Tunnel access is from the outside, or on the surface of a shaft when the configuration requires it. Any person entering the tunnel must be equipped with a counting system (RFID chip or equivalent).

When a turnstile or gate type access control is installed, this equipment has an unlocking box on the inside in the event of an emergency to allow evacuation and emergency access without obstacle.

NB: For staff access via mechanized means (TSP, van, etc.), each staff must wear their tunnel RFID chip if they have not passed through one of the access gates. External delivery drivers (concrete for example) must also be identifiable in the tunnel.

All people holding a tunnel access badge must be trained:

- Underground work for site personnel (supervisors, companions, subcontractors, etc.)
- Safety/evacuation instructions for visitors. They also must be accompanied by personnel who know and are trained in evacuation and emergency instructions.

RFID chip – 1 per person

RFID chip for visitors, subontractors mandatory inside the tunnel.

Tunnel access is managed by antennas locating the passage of the RFID chip and making it possible to count the number of people between 2 antennas. A surface/exterior server makes it possible to count personnel in real time, to edit a list of underground personnel and to update in real time the displays on the surface, entrance to underground zone, cave zone, and/or TBM.

Each shaft and tunnel section must be delimited by zone entry and exit detection antennas.

Particular attention should be paid:

- When cross passages are opened on shafts, or interconnected parallel tunnels, or existing tunnel, etc.
- When crossing stations, in coactivity with civil engineering personnel, who may not be equipped with the tunnel RFID chip.

The number of personnel authorized in the tunnel is determined according to the capacity of the refuge chamber(s), from the last open access from the tunnel to the outside.

For example, a digital display will be installed at the entrance to the tunnel, and another at the entrance to the area considered to be a cave (between the last open branch and the working face) or on the TBM, in order to indicate the number of people present.

The number of visitors authorized in the tunnel simultaneously is determined according to the project configurations, as per the precepts stated previously. Visitors must be accompanied by a project member, duly trained in evacuation and emergency procedures.

NB: During shift change, any visit to the tunnel is strictly prohibited.

4. Tunnel communication

Communications in the tunnel are primarily through radios/walkie-talkies and cell phones. The radios are previously defined in several groups (including a fire emergency channel).

In tunnels, emergency alarm and telephone boxes are installed from 200 m from the tunnel entrance, and at regular distances, depending on local, customer and emergency services recommendations. Without specific recommendations defined, these will be installed every 400m.

These boxes must remain accessible at all times:

- Do not store anything in front
- In the case of a suspended pedestrian bridge: from the bottom of the base using compliant means of access

Red light signage is placed above the box to make the nearest box visible.

These boxes can communicate via WiFi and include:

- Tunnel Alarm Function
- Communication function (telephone) + list of emergency numbers
- Lighting function
- Electrical distribution function
- First aid kit
- Fire extinguishers

5. Lighting and electrical network in tunnel

Depending on site configurations, some tunnel equipment powered by low voltage (LV) may be kept functional, for example (non-exhaustive list):

- Lighting
- Elevator
- Ventilation (generally 50% of power)
- Compressors, boosters
- Pumping system
- Electrical boxes and computer networks
- *Any equipment whose risk analysis/operating mode would make it obligatory to maintain it in minimum operation*

The High Voltage (HT) power supply comes from the outdoor delivery station and is relayed by the High Voltage departure station in the tunnel.
In the event of a fire, an emergency stop on the HV departure substation in the tunnel allows the distribution in the tunnel to be completely stopped, as well as on the equipment inside the tunnel.

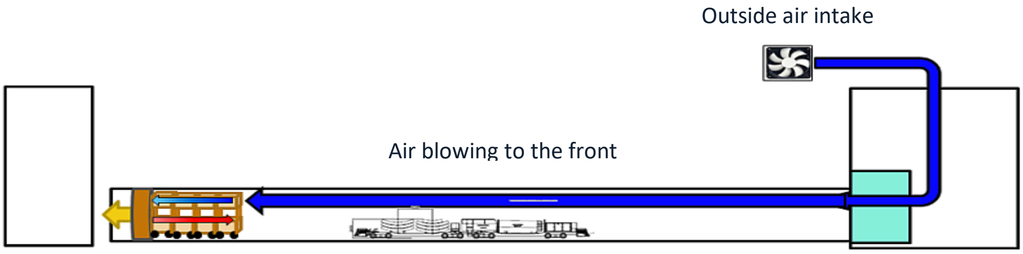
Special case of TBM : the equipment kept functional from a tunnel boring machine will be determined according to the operating mode of the machine (mortar agitator, refuge, erectors, computer network, pumping systems, etc.)

6. Tunnel ventilation

Tunnel ventilation is ensured by air blown from the outside through an outside air intake (allowing unpolluted air to be sent into the tunnel), from a rigid duct to the entrance of the tunnel, then a vent tube type duct. The sizing of the ventilation ducts will be determined in the preliminary ventilation study, taking into account the minimum air flow to be provided.

This dimensioning takes into account in particular the developments of the project with the integration of workshops complementary to the excavation: opening of the cross passages, rear workstations and associated constraints, dust in particular (raft, gravel-cement, installation of rails, etc.). All measures must be taken to ensure optimal operation in any underground location.

The air flow will be regularly monitored along the entire tunnel route.



Work premises installed in tunnels must also be the subject of a calculation note.

In the event of a power outage (except fire), the ventilation must be backed up at a minimum of 50% of its initial power.

In the event of a fire, tunnel ventilation will be cut off immediately. The ventilation emergency stop button must be easily accessible, located and known to everyone.

NB: The ventilation installation, once the excavation and civil engineering have been completed, must be adapted to the finishing work activities (customer discussion).

7. Tunnel water supply

One or more valves with firefighter connections at the entrance to the tunnel allow the immediate connection of a fire hose to the wet column on each of the equipped pipes.

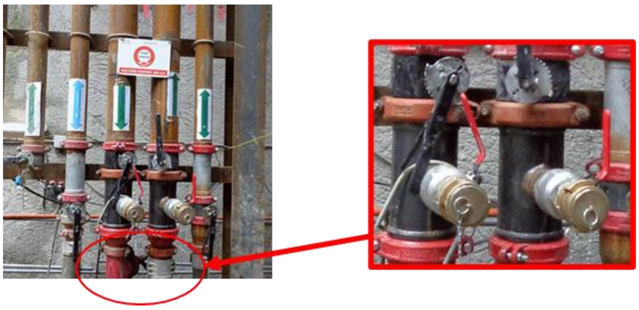
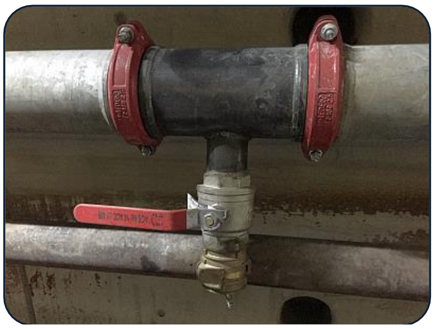


Photo of the location of the valve + firefighter connection in the well

The diameters of the fire brigade connections must correspond to the country's standards (ø65 mm for France for example), they are arranged according to local regulations, and at least every 200m, and can provide 7 Bars to 2 simultaneous lances. Firefighter connections must be easily accessible from the bottom of the tunnel and identified.



Valve with connection DN65

Identification of Firefighter valve

Note: If water lances are connected, favor consumption on the return column if personnel are on the TBM in order to preserve water resources for protection by a water curtain.

Depending on the configurations discussed with the emergency services, an on-site water reserve may be placed on the surface. This can be re-supplied directly by a pump machine via a fire hose.

Note: The water supply installation in the tunnel, once the excavation is completed, must be adapted to the finishing activities (customer discussion): dry or wet column with booster for example.

8. Hazardous products/waste and radioactive sources in tunnels

The storage of chemicals and/or dangerous waste, particularly flammable ones, is prohibited in tunnels. The reloading of vehicles with diesel, in shafts in particular, must be subject to a specific risk analysis and associated fire-fighting means.
Sites using Drill & Blast digging mode must comply with specific rules for storage, supply and retrieval of the explosives/emulsions used.
Special case of radioactive sources in tunnels (TBM slurry): The Competent Person in Radiation Protection must, in collaboration with the TBM site and H&S team, define the specific procedures for closing/managing radioactive sources in relation to the fire risk. Radioactive sources must be identified by specific display.

9. Hot work interventions in tunnel

Any worker wishing to carry out hot spot work (grinder, welding, ox cutting, etc.) in the following areas (non-exhaustive list) must first complete **a fire permit:**

- Tunnel
- Tunneling machine if applicable (unless exemption in the workshop area)
- Underpinning work areas
- Areas not naturally ventilated (confined spaces)
- ATEX defined zones

- Areas located near storage of chemicals, hazardous waste, gas cylinders
- Mechanical interventions by hot spot near fuel tanks
- ***All areas where the risk analysis would make it mandatory***

For these areas, it will be necessary to have, in addition to fire equipment already in place (on the TBM for example), one (or more) fire extinguisher(s) in the immediate vicinity of the hot work station.

Preparing the area before work using a hot spot is mandatory, as well as monitoring this same area for 2 hours following the end of the intervention (checking for the absence of smoldering fire for example), these points are covered in fire permits.

As a reminder, smoking is prohibited in confined spaces, in tunnels. A specific risk analysis will determine smoking areas, based on the activities and associated risks.

10. Tunnel emergency/fire equipment

10.1. Means of access and evacuation in tunnels

Pedestrian entrances to tunnels are differentiated from construction site tracks/machine circulation areas by:

- Physical marking at the entrance and at salient points (cross passages, vehicle turning zones, etc.), then specific markings at regular intervals
- A suspended walkway when the configuration requires it
- Any other system allowing the management of risks linked to coactivity between vehicles and pedestrians

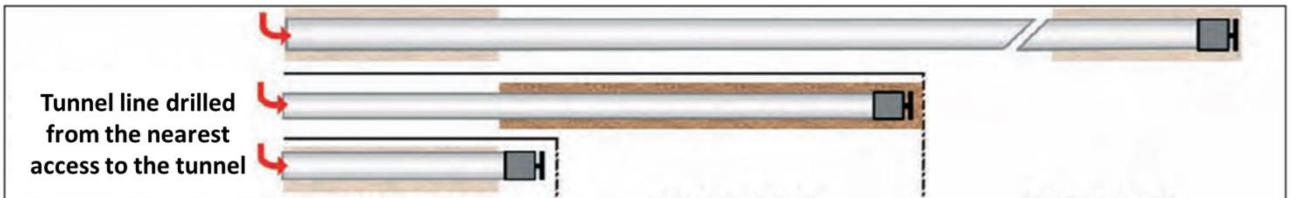
A counting system linked to the access badge and a display system for the number of people present in the tunnel at "t" time must be put in place, including in the supervision room.

For access to shafts, these are equipped with an Access tower or staircase type access means, accessible and in good condition for access and evacuation from the bottom of the shaft. Depending on the work phases, also depending on local regulations, elevators will be put in place (do not use the elevator in the event of evacuation) in the event of access to a shaft of depth >14m.

The use of a suspended manbasket from a lifting means is prohibited for lowering personnel (see **Standard Fall from height prevention [BYTP-REF-H&S-2211]**). If, for any technical reasons, or linked to the digging method (drill & blast for example), this has to be used, specific conditions of use will be implemented, and validated by the Project Management as well as the H&S Manager (technical impossibility of doing otherwise to demonstrate).

The main entrance to a tunnel is considered as the first emergency exit. As long as the configuration of the tunnel under construction does not include other emergency exits (cross passages, shafts, twin tunnel, etc.), the following measures must be taken:

Source AFTES booklet GT4R5F1:



	Configuration A $L_{Bored\ tunnel} \leqslant 200$ m	Configuration B $L_{Bored\ tunnel} \leqslant 500$ m	Configuration C $L_{Bored\ tunnel} > 500$ m
$Ø_{Interior} \geqslant 3,00$ m	APPLICATION OF PATHWAYS depending on the zone (free tunnel, follower train, transition zone, machine)	APPLICATION OF PATHWAYS depending on the zone (free tunnel, follower train, transition zone, machine) Autonomous protective equipment for the personnel	APPLICATION OF PATHWAYS depending on the zone (free tunnel, follower train, transition zone, machine) Autonomous protective equipment for the personnel SURVIVAL SHELTERS (on the machine and if required in the current area)
$2,20\ m < Ø_{Interior} < 3,00$ m	SPECIFIC BRAINSTORMING from stakeholders (Client, Project management, H&S Manager, emergency services) at the different design stages regarding safety during execution of the work: • Encourage the establishment of survival shelters, niches, etc. • Choose specific equipment (electrical for example) • Conditions on pedestrian traffic		• Multiply accesses to get out of case C • Favor the use of "unmanned" tunnel boring machines
$Ø_{Interior} \leqslant 2,20$ m	USE "UNMANNED" TUNNELIERS		

Configuration A: Linear tunnel from the nearest access point < 200m
Configuration B: Linear tunnel from the nearest access point < 500m.
If other work is carried out in addition to the digging, then the configuration passes to C (increased fire risk, due to rear activities)
Configuration C: The tunnel has at least one remote point with a length >500m, from the nearest usable access point. Throughout the tunnel, in addition to the means of autonomous protection, it is necessary to provide a survival shelter, either on the tunnel boring machine, or at the level of the working face (between 250 and 500m back from the face, depending on the activities), and if necessary, to the right of the sites annexed to that of the main excavation.

Depending on the progress of the work in the main tunnel, some cross passages may be opened and thus used as emergency exits but also as a preferred means of access for the emergency services.

- Using a branch only as an emergency exit:
  - Mandatory smoke protection, 30 min fire break, to avoid contamination of evacuation areas by fire smoke
  - Specific green LED lighting backed up
  - Personal counting antenna at the right of the open branch
  - Self-rescue masks if applicable
  - *Any other necessary or imposed provisions*
- Use *of a branch as an emergency exit **AND** firefighter access:*
  - Mandatory 30min fire-rated SAS with 2 30min fire-rated doors, both equipped with a door closer, these opening inwards
  - Doors with a minimum width of 0.90m and a minimum height of 2.00m, with a 15cm recess in the lower part
  - SAS carried out as close as possible to the tunnel outlet
  - Non-pressurized airlock

- Installation of a rigid or semi-rigid pipe supplied from outside the structure by a firefighter connection, allowing firefighters to access the tunnel under the protection of hydraulic lances under load
- No material storage or electrical cabinet in the SAS

All of these provisions must be integrated from the design phase and presented to the emergency services.

Some special cases will have to be considered: coactivity with a tunnel in operation, under circulation, case of twin tunnels connected by branches, etc.

10.2. Emergency equipment at tunnel/shaft entrance

An emergency cabinet is available at the bottom of the shaft or at the tunnel entrance. It has a fire extinguisher, a first aid kit and a portable safety shower.

A "fire box" is also located near the emergency cabinet. This has:

- Walkie-talkies
- Tunnel access badges
- Up-to-date plans of the tunnel, open branches…
- List of Local Incident Correspondent and Principal Incident Correspondent
- List of important phone numbers
- Local Incident Correspondent, Principal Incident Correspondent and Crisis Management Helper retroreflective vests
- Flashlights
- etc.

For configurations with access via shafts, an emergency basket is also available on the right-of-way (grabbing by the lifting means on site), and made available to emergency services in the event of necessary evacuation of a horizontal victim. .

This emergency manbasket will be deployed at the request of emergency services or in the event of serious and imminent danger requiring the immediate removal of personnel(s) or victim(s) from a risk area.

The ventilation stop buttons and evacuation sound alarm are also marked and available.

10.3. Construction machinery entering a tunnel

Construction equipment is equipped with fire-fighting equipment such as:

- Automatic shutdown on engine parts, tires, etc.
- Manual extinguishing means (fire extinguishers)

A specific risk analysis by type of machine will determine those which will be equipped with automatic extinguishing in the event of fire.
TSP type construction machinery, construction vans, under conditions permitting, can be used to evacuate personnel more quickly.

10.4. Tunnel boring machine

By design, the tunnel boring machines will be equipped, among other things:

- Sprinklers
- Of water curtains
- Gas detection
- Fire detection
- Survival shelter
- Self-rescue masks
- Fire extinguishers
- armed fire taps
- ...

The fire defense plans must be provided by the tunnel boring machine supplier, and tested on site during Commissioning .

10.5. Refuge chambers

Refuge chambers are artificial shelters allowing tunnel personnel to regroup to wait for help. These survival shelters must only be used as a last resort, the priority instruction, in the event of invasion of the tunnel **by smoke or harmful gases** , being evacuation to the outside.

From 500m in the cave in the tunnel, the establishment of refuge chamber(s) is obligatory, regardless of the digging method. Their number must be calculated based on the number of people present in the tunnel, and the progress of the work (blind window, opening of branches, etc.).

A shelter **should not be** used as direct protection against:

- A risk of falling rubble/rocks
- Overpressure blasting
- A fire
- The presence of flammable products
- Flooding...

It should be noted, as previously stated, that the number of places in the refuge chamber determines the number of tunnel entries.

Refuge chambers are equipped with specific boxes including: drinking water, lamps, wipes, tissues, survival blankets, energy bars, card games, etc.

Autonomy:

- Is increased by the arrival of compressed air from the external network
- Starts only when opening oxygen cylinders and closing compressed air/breaking surface compressed air supply

Specific training in the management and use of survival shelters is mandatory for people working in tunnels.

10.6. Specific PPE

Personnel entering the tunnel must, in addition to the standard PPE required at all times, wear:

- A helmet headlamp (in case of invisibility due to fire smoke for example)
- A RFID chip control badge
- Permanently reflective clothing

10.7. Self-rescue masks

The use of self-rescue masks may be necessary in the event of fire and/or poor atmosphere in a tunnel.

In the event of an evacuation or alarm, staff are instructed to retrieve a self-rescue mask for evacuation. The mask can then be used upon contact with fire or smoke.

Self-rescue masks must be sized according to the most unfavorable route to be covered on foot in the event of an emergency along the route of the tunnel to be created.
User training is mandatory for personnel working in tunnels. These training courses are provided by qualified personnel. Underground visit supervisors must also be trained in the use of these self-rescue masks.

The location of these must be known to personnel working in the tunnel. They are generally positioned:

- In the refuge chambers
- At the salient points of the routes (branches)
- In construction machinery / construction vans (or MSV)

Example: OCENCO EBA 6.5 self-rescue masks

Autonomy:
- 90 minutes in evacuation
- Mask with compressed air cartridge

Self-contained closed circuit breathing apparatus
Automatic priming by breath

10.8. Other tunnel rescue equipment

A defibrillator (AED) is available in the refuge chamber or in a defined construction site area known to all

First aid kits are available in refuge chambers, evacuation SAS

Eye washes are available with each first aid kit.

A portable safety shower is available in the refuge chamber and/or in addition to other emergency equipment.

A fully restrained stretcher that can be lifted is available in the refuge chamber or any other area consistent with the activities in progress.

It is also available for any emergency rescue/release and is suitable for rescue in the slaughter chamber and vertical handling.

11. Tunnel Emergency Response Team

Each construction site ensures the establishment of an Emergency Response Team responsible for managing any potential emergency and establishes an Emergency Management Center, which will serve as the operational center for the emergency team.

11.1. Command post

In the event of a serious event requiring the intervention of the emergency services, a room permanently equipped with up-to-date maps/plans/synoptics and transmission equipment (telephone/video/internet, etc.) must be immediately available for the management of relief operations.

This room can accommodate at least 15 people and is called “Command Post" (CP). Ideally, this room is on the ground floor, easily accessible, and close to any digging monitoring rooms (tunnel supervision for example)

The Command Post is equipped:

- Means of communication specific to crisis management (radios with specific lines)
- A whiteboard with markers of different colors
- A plan of the tunnel making it possible to locate the position of the workstations on the route and the possible workstations at the rear of the tunnel (the same for a tunnel boring site)
- Display screens linked to access controls and RFID badges
- Identification vests for people with a specific role in crisis management (Local Incident Correspondent, Principal Incident Correspondent and Crisis Management Helper)
- Up-to-date organization charts of the teams on duty with the numbers of the managers to contact by team.

11.2. Definition of roles in the event of a crisis

Only internal contacts likely to be physically and organizationally integrated into ongoing crisis management are defined in this procedure. The crisis management procedure linked to customer interfaces, project management, residents, etc. is not covered here.

The responders named below LIC, PIC, and CMH will be clearly identified to the emergency services and are identifiable by wearing a specific retroreflective identification chasuble.

The Principal Incident Correspondent (PIC):

- Is responsible for managing Crisis situations
- Activates the Central Emergency Team
- Must be able to arrive on site as quickly as possible and, in any case, in less than an hour
- Ensures the interface between the site/project and the command of external intervention services
- Has all the elements concerning the work in progress
- Is able to take all necessary measures for the smooth running of relief operations
- Notifies its operational entity and keeps it regularly informed of developments in the situation
- Ensures effective communication with authorities, the media, and the public

The Local Incident Correspondent (LIC):

The LIC, designated by the Delegate of Power/Project Manager, is a trained member of the works management, having the material and technical knowledge necessary for emergency management on site.

He is permanently present on site and:

- Ensures logistics related to emergency management
- Supervises a team whose role is to ensure the guidance of rescue operations and logistical support for rescue personnel (lifting, transport of personnel and/or equipment, etc.)
- Is the on-site contact person for external intervention services
- Allows on-site access for external intervention services
- Informs the emergency services of the situation of dangerous networks and products
- Has full authority over all staff present, whatever the hierarchical level and/or the company they belong to (within the framework of their emergency management mission)

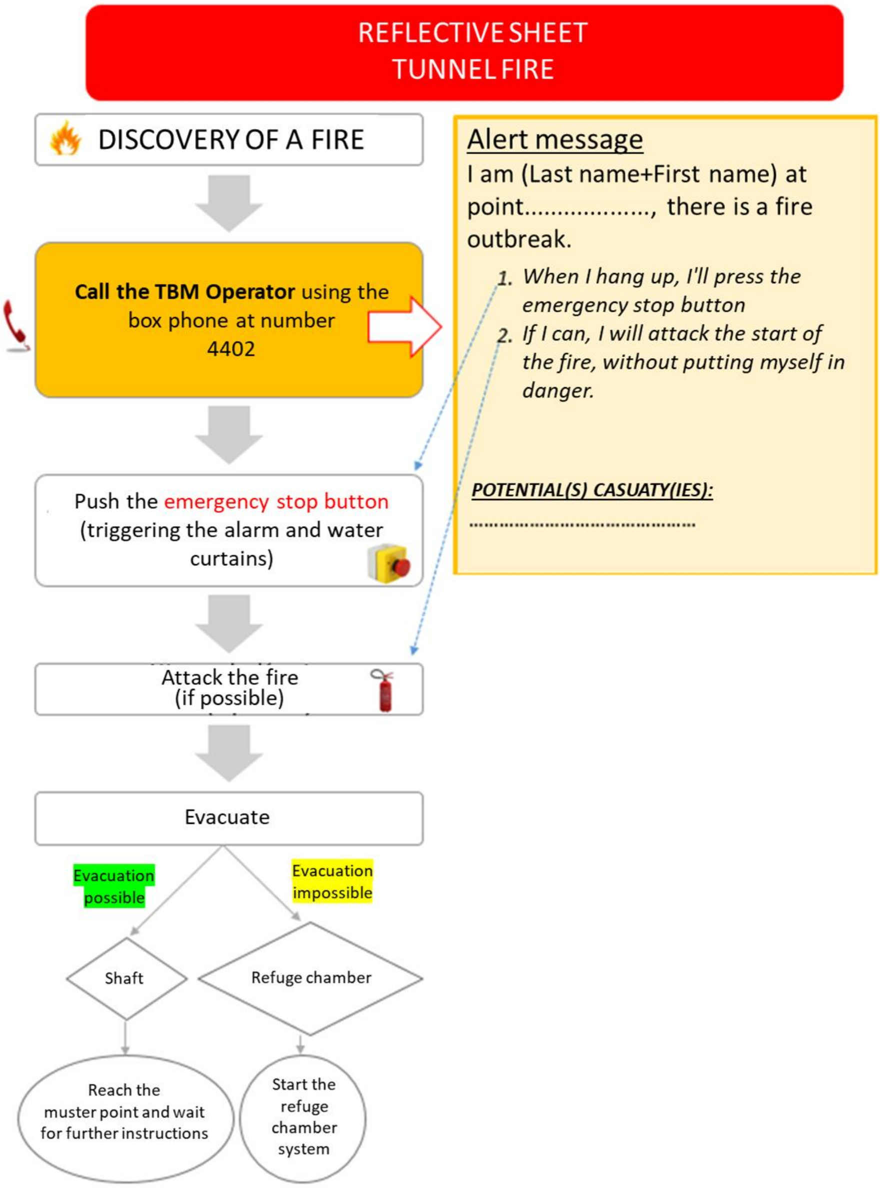
In addition to the LIC and PIC:

Crisis Management Helper (CMH):

Under the responsibility of the CIL, the CMH knows perfectly the configuration of the site(s) affected by the current crisis:

- He helps with the installation of markings and access to firefighters
- It opens the gates/accesses
- He knows how to use the gantry crane if necessary (Training + Driving authorisation), or asks for somebody who knows
- etc.

11.3. Example of tunnel fire reflex sheet



11.4. Tunnel fire reflex sheet

Principal Incident Correspondent - Checklist

Name + Tel Number:

Date/Hour:

Before emergency services arrival	Status
Prepare the crisis management unit	
Ensure that preparations for the intervention have been done by the LIC	
Cross-reference information on pesonnel counting via the supervision room	
Indicates on the map the position of (potentiel) blocked people	
Identifies the position of the TBM on the map	
Transmit emergency radios located in the control room on the site	
Lists all the elements on the work in progress and the technical means available for the intervention (crane, gantry trolleys, etc.)	

During the intervention	Status
Ensure the transmission of information to the client and take charge of communication/information to the authorities, media and public if necessary	

Local Incident Correspondent - Checklist

Name + Tel Number:

Date/Hour:

Before emergency services arrival	Status
Ensure that the emergency services are informed of the emergency	
Ensure that the emergency alarm had been raised	
Ensure that the evacuation/mustering of personnel is performed	
Collect the figures of personnel still in the tunnel	
Ensure that the ventilation is stopped	
Ensure that the worksite entrance ar e opened for the emergency services	
Ensure that the defined area of the worksite is free for the mergency services to be deployed at their arrival	
Ensure that the crisis management unit is ready (Tunnel and TBM plans available, acces indication in the site installation plan, etc.)	
Ensure that the crane is equipped with an emergency personnel transport basket	
Ensure that sling are available to lift any required additional material	

At the emergency services arrival	Status
Ensure that a CMH is guiding the emergency services from the site entrance to the emergency crisis room	
Ensure that they are receiving the local radio with the secour channel already set-up	
Provide them with the information on the type and location of the incident	
Provide htem with the location of the water supply sources available	
Ensure that they are informed of the number and known localisation of personnel in the tunnel	
Inform them with the first actions performed	
Ensure that they are informed of the closed access to the incident	

During the intervention	Status
Remain available at all time	
At the PIC arrival, go to the forward command post and remain avaiable to support the emergency services without been physically involved	